

Continuity & Measurability of Multifunctions

Pathak Chapter 10b: Applications to Integral Inclusions

H. K. Pathak

Nonlinear Analysis and Fixed Point Theory

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Chapter Overview

- **Topic:** Fixed point theorems for multifunctions and their applications
- **Focus:** Continuous and measurable multifunctions
- **Applications:** Nonlinear integral inclusions and differential inclusions
- **Key Concepts:**
 - 1 Continuity properties of set-valued maps
 - 2 Measurability of multifunctions
 - 3 Fixed point theorems
 - 4 Integral inclusions
 - 5 Hammerstein type equations

1. Introduction to Multifunctions I

Definition: A multifunction (set-valued map) $F : X \rightarrow 2^Y$ assigns to each $x \in X$ a set $F(x) \subseteq Y$.

Key Distinctions:

- Single-valued functions map points to points
- Multifunctions map points to sets
- Can express inclusion problems elegantly

Examples of Multifunctions:

- 1 Normal cone to a convex set
- 2 Subdifferential of a convex function: $\partial f(x) = \{y \in X^* : \langle y, z - x \rangle \leq f(z) - f(x) \text{ for all } z\}$
- 3 Solutions set of differential inclusions

Motivation:

- Model uncertainty and set-valued constraints
- Study existence of solutions to inclusions rather than equations
- Unified framework for many problems

2. Continuity of Multifunctions I

Definition (Upper Semicontinuity): A multifunction $F : X \rightarrow 2^Y$ is upper semicontinuous (u.s.c.) at x_0 if for every open set U containing $F(x_0)$, there exists a neighborhood V of x_0 such that $F(x) \subseteq U$ for all $x \in V$.

Definition (Lower Semicontinuity): F is lower semicontinuous (l.s.c.) at x_0 if for every open set U intersecting $F(x_0)$, there exists a neighborhood V of x_0 such that $F(x) \cap U \neq \emptyset$ for all $x \in V$.

Complete Continuity: F is completely continuous if it maps bounded sets to sets with compact closure.

Key Properties:

- **Closed-valuedness:** F is u.s.c. and locally bounded $\Rightarrow F$ is closed-valued
- **Compactness:** If X is compact and F is u.s.c. with compact values $\Rightarrow$ graph of F is compact
- **Continuity:** F is continuous iff both u.s.c. and l.s.c.

3. Measurability of Multifunctions I

Definition: A multifunction $F : \Omega \rightarrow 2^X$ (where Ω is a measure space) is measurable if:

$$F^-(B) := \{s \in \Omega : F(s) \cap B \neq \emptyset\} \text{ is measurable for all open } B \subseteq X$$

Equivalently:

$$F^+(K) := \{s \in \Omega : F(s) \subseteq K\} \text{ is measurable for all compact } K \subseteq X$$

Measurable Selection Theorem: If $F : \Omega \rightarrow 2^X$ is measurable with nonempty closed values, then there exists a measurable selection $f : \Omega \rightarrow X$ such that $f(s) \in F(s)$ for almost all $s \in \Omega$.

Practical Importance:

- Allows integration of set-valued maps via selections
- Essential for variational methods
- Links multifunctions to classical measure theory

4. Fixed Point Theorems for Multifunctions I

Theorem 5.167 (Kakutani-Fan-Glicksberg): Let K be a nonempty compact convex subset of a Banach space X . If $F : K \rightarrow 2^K$ is u.s.c. with nonempty compact convex values, then F has a fixed point, i.e., there exists $x \in K$ such that $x \in F(x)$.

Important Consequence: For applications to integral inclusions, we can construct operator $A : C \rightarrow 2^C$ (or $A : C \rightarrow K(C)$) such that fixed points correspond to solutions.

Completeness Conditions:

- F is upper semicontinuous
- F maps into nonempty compact convex sets
- Domain is nonempty compact convex subset of Banach space

Application Strategy:

- 1 Formulate inclusion as multifunction equation
- 2 Verify continuity and convexity conditions

4. Fixed Point Theorems for Multifunctions II

- ③ Apply fixed point theorem
- ④ Construct solution from fixed point

5. Nonlinear Integral Inclusions I

General Form:

$$x(t) \in \lambda(t) + \int_0^t [a(s, x(s)) + f(s, x(s), u(s))] ds$$

for $t \in [0, T]$, where:

- $\lambda \in C[0, T]$ is given
- $a : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$
- $f : [0, T] \times \mathbb{R}^2 \rightarrow 2^{\mathbb{R}}$ is a multifunction
- $u : [0, T] \rightarrow \mathbb{R}$ is a measurable function

5. Nonlinear Integral Inclusions II

Solution Space: $E = C([0, T], \mathbb{R})$ with the supremum norm $\|x\|_\infty = \sup_{t \in [0, T]} |x(t)|$.

Operator Formulation: Let $A : C \rightarrow 2^C$ be defined by:

$$Ax = \left\{ y \in C[0, T] : y(t) \in \lambda(t) + \int_0^t [a(s, x(s)) + f(s, x(s), u(s))] ds \right\}$$

Solution: x is a solution if $x \in Ax$ (fixed point of A).

6. Continuous Solutions (Section 10.1) I

Problem 10.1: Find continuous solutions to nonlinear integral inclusions

Theorem (Continuous Solution Existence): Consider the inclusion

$$x(t) \in \lambda(t) + \int_0^t [a(s, x(s)) + f(s, x(s), u(s))] ds, \quad t \in [0, T]$$

Hypotheses:

- ① $\lambda \in C[0, T]$, a, f satisfy Carathéodory conditions
- ② Appropriate growth conditions: $|a(s, x)| \leq k_1(s)$, $|f(s, x)| \leq k_2(s)$
- ③ Measurable selections exist for f
- ④ Upper semicontinuity conditions on the operator

Key Steps:

- Construct solution set in bounded subset $C \subseteq C[0, T]$
- Apply Kakutani fixed point theorem

6. Continuous Solutions (Section 10.1) II

- Use Nemytskii operator properties
- Verify multifunction has fixed point

Result: Under suitable conditions, there exists at least one continuous solution.

7. Nemytskii Operator I

Definition: Let $f : \Omega \times \mathbb{R}^n \rightarrow \mathbb{R}$ be a Carathéodory function (i.e., continuous in x for a.e. s , measurable in s for all x). The Nemytskii operator is:

$$N_f(u)(s) = f(s, u(s))$$

Theorem 1.42 (Nemytskii Property): If $\mu(\Omega) < \infty$ and f is Carathéodory, then:
If $\{x_k(s)\}$ converges to $x_0(s)$ in measure, then $\{N_f x_k(s)\}$ converges to $N_f x_0(s)$ in measure.

Continuity Theorem: Suppose N_f maps L_p into L_q (where $\frac{1}{p} + \frac{1}{q} = 1$). Then:

N_f is continuous and bounded.

Applications:

- Validates that solution operators are well-defined
- Ensures compositions of operators preserve measurability
- Critical for weak solutions in PDE theory

8. Hammerstein Integral Inclusions (Section 10.2) I

Form:

$$x(t) \in \lambda(t) + \int_0^t k(t, s)g(s, x(s))ds$$

where $k : [0, T]^2 \rightarrow \mathbb{R}$ is a kernel, $g : [0, T] \times \mathbb{R} \rightarrow 2^{\mathbb{R}}$ is a multifunction.

Key Assumptions:

- 1 Kernel $k(t, s)$ satisfies uniform boundedness:

$$\sup_{t \in [0, T]} \int_0^T |k(t, s)| ds < \infty$$

- 2 Multifunction g has measurable selections
- 3 Growth condition: $|g(s, x)| \leq M|x| + N$ for some constants

8. Hammerstein Integral Inclusions (Section 10.2) II

Existence Theorem: Under the above hypotheses and contractivity conditions, there exists a continuous solution.

Proof Strategy:

- Construct operator $M_{x,\sigma}(t) := F(t, V(x_{\sigma,\lambda}(t)))$
- Define selection operators $T(\sigma)$
- Verify contractivity via Hypothesis 10.2
- Apply Banach fixed point theorem

9. Filippov Type Existence Theorem (Section 10.3) I

Problem: Prove existence for nonconvex integral inclusions using Filippov approach.

Setup:

$$x(t) \in \lambda(t) + \int_0^t [a(t, s)g(t, s(u(s)) + f(t, s, u(s))]ds$$

where a, g, f are given and may be nonconvex.

Key Theorem:

- ① Construct auxiliary set-valued maps $\tilde{F}$ and $\tilde{T}$
- ② Verify measurability with nonempty closed values
- ③ Show $\tilde{T}(\sigma)$ is a contraction in L_1 :

$$H^+(T_\sigma(\sigma_1), T_\sigma(\sigma_2)) \leq \frac{1}{\sqrt{\alpha}} \|\sigma_1 - \sigma_2\|_1$$

- ④ Apply Banach contraction principle

9. Filippov Type Existence Theorem (Section 10.3) II

Important Features:

- Works for nonconvex-valued multifunctions
- Combines measurable selection theory with contraction principle
- Exponential weighting ensures contractivity

10. Numerical Example: Integral Inclusion

Problem: Solve numerically

$$x(t) \in 0.5 + \int_0^t 0.1 \sin(s) x(s) ds$$

on $[0, 1]$ using piecewise linear approximations.

```
1 import numpy as np
2 from scipy.integrate import odeint
3 import matplotlib.pyplot as plt
4
5 def integrand(x, t):
6     """Approximate the kernel integral"""
7     return 0.1 * np.sin(t) * x
8
9 def solve_inclusion():
10     # Initial condition and time points
11     x0 = 0.5
12     t = np.linspace(0, 1, 100)
13
14     # Solve ODE: dx/dt = 0.1*sin(t)*x
15     # Solution: x(t) = x0 * exp(integral)
16     solution = odeint(integrand, x0, t)
17
18     return t, solution.flatten()
19
20 t, x = solve_inclusion()
21 print(f"Solution at t=1: {x[-1]:.6f}")
```

11. Numerical Implementation (Continued)

```
1 def verify_solution(t, x):
2     """Verify the integral inclusion"""
3     dt = t[1] - t[0]
4
5     # Compute integral term
6     integrand_vals = 0.1 * np.sin(t[:-1]) * x[:-1]
7     integral = np.cumsum(integrand_vals) * dt
8
9     # Compute inclusion RHS
10    rhs = 0.5 + integral
11
12    # Residual
13    error = np.abs(x[1:] - rhs)
14
15    return error.max(), error
16
17 # Verify and plot
18 error_max, error = verify_solution(t, x)
19 print(f"Maximum error: {error_max:.2e}")
20
21 plt.figure(figsize=(10, 6))
22 plt.plot(t, x, 'b-', linewidth=2, label='Solution')
23 plt.grid(True, alpha=0.3)
24 plt.xlabel('Time t')
25 plt.ylabel('x(t)')
26 plt.title('Integral Inclusion Solution')
27 plt.legend()
```

12. Key Theorems Summary I

Theorem 1: (Upper Semicontinuity and Complete Continuity)

If $A : C \rightarrow K(C)$ is u.s.c. and completely continuous (maps bounded sets to precompact sets), then A has a fixed point.

Theorem 2: (Measurable Selections)

If $F : \Omega \rightarrow 2^X$ is measurable with nonempty closed values, there exists a measurable selection $f : \Omega \rightarrow X$ with $f(s) \in F(s)$ a.e.

Theorem 3: (Nemytskii Operator Boundedness)

Under Carathéodory conditions with appropriate growth, the Nemytskii operator is continuous and bounded.

Theorem 4: (Contraction Principle for Multifunctions)

If $T : L_1 \rightarrow 2^{L_1}$ is a contraction in the Hausdorff metric with compression ratio $\rho < 1$, then T has a unique fixed point.

13. Key Definitions and Notation I

Standard Spaces:

- $C[0, T]$: Continuous functions on $[0, T]$ with $\|x\|_\infty = \sup_t |x(t)|$
- $L_p[0, T]$: Lebesgue integrable functions with $\|x\|_p = \left(\int_0^T |x(t)|^p dt \right)^{1/p}$
- 2^X : Power set of X (all subsets)
- $K(C)$: Family of nonempty compact subsets of C

Multifunction Properties:

- U.S.C. (upper semicontinuous): Inverse images of open sets are open
- L.S.C. (lower semicontinuous): For open U intersecting $F(x)$, nearby points have image intersecting U
- Measurable: Preimage of open set is measurable

Distances:

- Hausdorff distance: $H(A, B) = \max\{\sup_{a \in A} d(a, B), \sup_{b \in B} d(b, A)\}$
- Used to measure distance between sets

14. Applications and Extensions I

ODE Applications:

- Differential inclusions: $x'(t) \in F(t, x(t))$
- Uncertain differential equations
- Optimal control with set-valued constraints

PDE Applications:

- Parabolic differential inclusions
- Evolution inclusions with monotone operators
- Systems with hysteresis effects

Advanced Topics:

- Parametric inclusions and sensitivity analysis
- Viability theory (solutions staying in given set)
- Multivalued stochastic differential equations

14. Applications and Extensions II

Practical Uses:

- Control theory under uncertainty
- Variational inequalities
- Image processing and optimization

15. Summary and Conclusions I

Main Concepts Covered:

- 1 **Multifunctions:** Set-valued maps generalize single-valued functions
- 2 **Continuity:** Upper and lower semicontinuity are key topological properties
- 3 **Measurability:** Essential for integrating multifunctions
- 4 **Fixed Points:** Kakutani-Fan-Glicksberg theorem applies to u.s.c. maps
- 5 **Integral Inclusions:** Applications to nonlinear equations with uncertainty

Key Theoretical Results:

- Existence of continuous solutions to integral inclusions
- Hammerstein type existence theorems
- Filippov's approach for nonconvex multifunctions
- Integration of multifunctions via selections

Why This Matters:

15. Summary and Conclusions II

- Provides rigorous framework for set-valued problems
- Connects topology, measure theory, and functional analysis
- Essential for modern applications in control and optimization

References and Further Study

Key References:

- ① Pathak, H.K. (2018). *An Introduction to Nonlinear Analysis and Fixed Point Theory*. Springer.
- ② Aubin, J.P. and Frankowska, H. (1990). *Set-Valued Analysis*. Birkhauser.
- ③ Deimling, K. (1992). *Multivalued Differential Equations*. De Gruyter.
- ④ Castaing, C. and Valadier, M. (1977). *Convex Analysis and Measurable Multifunctions*. Springer.

Topics for Further Exploration:

- Sensitivity analysis for parametric inclusions
- Fuzzy differential equations
- Stochastic multivalued equations
- Applications to game theory and economics